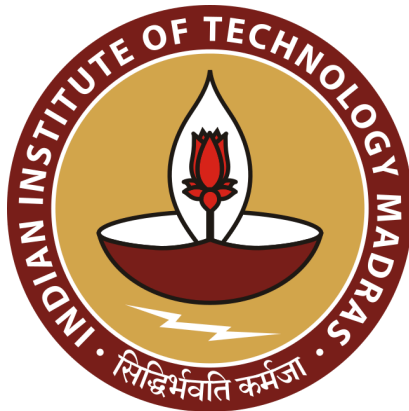


SYNTHESIS AND ANALYSIS OF THEO JANSEN MECHANISM

ME6221: THEORY OF MECHANISMS

**Jan-May 2024
Course Project**

**Under the Guidance of
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SYNTHESIS & ANALYSIS OF THEO JANSEN MECHANISM

1. ABSTRACT:

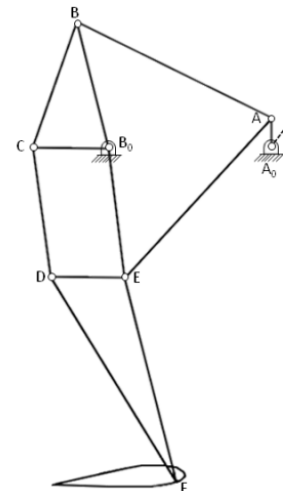
This project report aims to replicate and analyse the Theo Jansen mechanism, a renowned walking mechanism with only one DOF. This report will be discussing the method of synthesis using 3 precision points to achieve a natural walking motion and the kinematic & dynamic analysis using vector loop & force equilibrium respectively. This paper also contains the verification of the theoretical values found using simulation software and the design & fabrication of the final model.

2. INTRODUCTION:

Legged, off-road vehicles have proven superior in mobility, energy efficiency, and comfort on rough terrain compared to tracked or wheeled vehicles. Various leg mechanisms have been studied for applications including planetary exploration, disabled mobility aids, military transport, and rescue operations in hazardous environments. However, selecting and dimensioning leg mechanisms is complex due to their numerous links required for high mobility. This project aims to develop walking mechanisms inspired by natural creature movements, like the Jansen mechanism, focusing on accurate replication. Furthermore, the mechanism requires very simple controls since a single actuator is required to drive the leg.

Previous research in this field commonly emphasised two main approaches. Firstly, optimization methods focused on directly adjusting the length of each link, often requiring extensive time and effort due to the absence of structured procedures beyond altering link lengths and observing their effects on outcomes. Secondly, static analyses were employed to assess the forces acting on both links and joints, along with the torque applied to the crankshaft not taking into account the masses of the links.

Our objective is to analyse & synthesise mechanisms that accurately replicate this movement. We'll utilise techniques to create linkages that mimic ankle paths, analyse motion using MATLAB, and validate designs through CAD models and simulations. Fabrication will include 3D printing, followed by prototype assembly. Our motivation lies in the potential for advancements in locomotion systems for which the Jansen mechanism offers adaptability and efficiency in traversing various terrains.



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3. SYNTHESIS:

The synthesis of the mechanism consists of four parts - function generation of the input loop, coupler design, dyad synthesis and triad synthesis of the final loop. For simplification of notations, the following notations will be used.

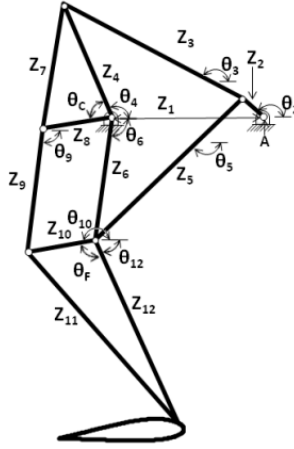


Figure 3.1- Labels for all links in the mechanism

3.1 3-Position Function Generation

The input loop $Z_1Z_2Z_3Z_4$ and $Z_1Z_2Z_5Z_6$ are treated as a function generation. The relationship between the input and output links is given as $y = A\sin(x-B) + C$, where $x_0 \leq x \leq x_{n+1}$, to mimic the walking motion of hip joints in humans with minimal vertical movement. The values of A, B and C are initialised as 30° , 65° and 105° respectively. ϕ represents θ_2 in the synthesis equations and ψ represents θ_4 or θ_6 i.e, input and output angles respectively. Φ and ψ assume a linear relationship with x and y respectively.

3.1.1 Chebyshev Spacing

The selection of precision points affects the stability of the solution and the structural error present in the mechanism. Chebyshev spacing provides a good approximation with minimal structural error. For n precision points, the Chebyshev spacing is defined as follows,

$$x_i = \frac{1}{2}(x_{n+1} + x_0) - \frac{1}{2}(x_{n+1} - x_0) \cos \frac{(2i-1)\pi}{2n}, \quad i = 1, 2, \dots, n$$

Using the equation $y = A\sin(x-B) + C$, precision points x_i and y_i are calculated and consequently the specific range of the input link is calculated. Assuming a linear relationship between Φ , ψ and x, y; the following equations give the precision points.

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$$\phi_i = \frac{\phi_{n+1} - \phi_0}{x_{n+1} - x_0}(x_i - x_0) + \phi_0, \quad i = 1, 2, \dots, n$$

$$\psi_i = \frac{\psi_{n+1} - \psi_0}{y_{n+1} - y_0}(y_i - y_0) + \psi_0, \quad i = 1, 2, \dots, n$$

From trial and error the free choice of the input angle range of complete 360° gave the most accurate solution. This gave the precision points of Φ as 24.1154° , 180.0000° , 335.8846° and of ψ as 85.3639° , 132.1892° , 75.0036° . These calculated precision points can be used with Freudenstein's method to calculate the link lengths of the input loop.

3.2 Freudenstein's Method

Freudenstein's equation relates the input link to the output link of a 4-bar mechanism which is used in the function generation of the said mechanism. With the input and output angles calculated in the previous section, a loop closure equation is set up and solved for the link lengths. Three equations for every position can be concisely written as

$$K_1 \cos \phi_1 + K_2 \cos \psi_1 + K_3 = -\cos (\phi_1 - \psi_1)$$

$$K_1 \cos \phi_2 + K_2 \cos \psi_2 + K_3 = -\cos (\phi_2 - \psi_2)$$

$$K_1 \cos \phi_3 + K_2 \cos \psi_3 + K_3 = -\cos (\phi_3 - \psi_3)$$

Where, $Z_2 = Z_1/K_2$, $Z_4 = Z_1/K_4$, $Z_3 = (Z_1^2 + Z_2^2 + Z_4^2 - 2K_3Z_2Z_4)^{1/2}$

With the free choice of the length of the ground link $Z_1 = 15\text{cm}$, the link lengths are calculated as $Z_2 = 4.17\text{cm}$, $Z_3 = 20.33\text{cm}$ and $Z_4 = 12\text{cm}$.

3.3 Coupler Design

The Freudenstein's method provides us with only Z_4 out of the coupler $Z_4Z_7Z_8$. Therefore the angle between Z_4 and Z_8 (θ_c) and the link length Z_8 are free choices. However, if the choice of θ_c is too large, the parallelogram will jam; or if it is too small, the parallelogram will reach a dead centre. Thus, the acceptable range satisfies the following conditions,

$$\theta_{cmax} = \angle BB_0E_{min}$$

$$\text{if } \angle BB_0E_{max} > 180^\circ \text{ then } \theta_{cmin} = \angle BB_0E_{max} - 180^\circ$$

$$\text{if } \angle BB_0E_{max} < 180^\circ \text{ then } \theta_{cmin} = 0^\circ$$

If $\theta_{cmin} > \theta_{cmax}$, then the mechanism will not function. Thus, θ_c was found out to be 60° .

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3.4 Dyad Synthesis

The dyad Z_6Z_{12} is synthesised by solving the following complex equations according to Burmester Theory.

$$\begin{bmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 \\ e^{i\beta_4} - 1 & e^{i\alpha_4} - 1 \end{bmatrix} \begin{bmatrix} Z_6 \\ Z_{12} \end{bmatrix} = \begin{bmatrix} \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix}$$

where β is the change in rotation of the link Z_6 , α is the change in rotation of link Z_{12} (free choices) and δ is the displacement vector of the end point at every precision point. The free choice β_2 is chosen to be -16.3° from trial and error. β_3 and β_4 are calculated from the following compatibility equations.

$$\Delta_2 = \begin{vmatrix} e^{i\alpha_3} - 1 & \delta_3 \\ e^{i\alpha_4} - 1 & \delta_4 \end{vmatrix}$$

$$\Delta_3 = - \begin{vmatrix} e^{i\alpha_2} - 1 & \delta_2 \\ e^{i\alpha_4} - 1 & \delta_4 \end{vmatrix}$$

$$\Delta_4 = \begin{vmatrix} e^{i\alpha_2} - 1 & \delta_2 \\ e^{i\alpha_3} - 1 & \delta_3 \end{vmatrix}$$

$$\Delta_1 = -\Delta_2 - \Delta_3 - \Delta_4$$

$$\Delta = \Delta_1 + \Delta_2 e^{i\beta_2}$$

$$\cos \theta_3 = \frac{|\Delta_4|^2 - |\Delta_3|^2 - |\Delta|^2}{|2\Delta_3\Delta|}$$

$$\sin \theta_3 = |(1 - \cos^2 \theta_3)^{1/2}| \geq 0$$

$$\theta_3 = \tan^{-1} \frac{\sin \theta_3}{\cos \theta_3}$$

$$\beta_3 = \arg \Delta + \theta_3 - \arg \Delta_3$$

$$\cos \theta_4 = \frac{|\Delta_3|^2 - |\Delta_4|^2 - |\Delta|^2}{|2\Delta_4\Delta|}$$

$$\sin \theta_4 = |(1 - \cos^2 \theta_4)^{1/2}| \geq 0$$

$$\theta_4 = \tan^{-1} \frac{\sin \theta_4}{\cos \theta_4}$$

$$\beta_4 = \arg \Delta + \theta_4 - \arg \Delta_4$$

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β_3 and β_4 are calculated as 68.7489° and 155.2678° by solving the above equations. Substituting these values in the first equation we get the link lengths Z_6 and Z_{12} to be 12 cm and 19 cm respectively.

3.5 Triad Synthesis

The synthesis of the triad $Z_8Z_9Z_{11}$ is a slight modification to the previous method where Z_8 is known. To calculate the link lengths Z_9 and Z_{11} , the following complex equations are solved.

$$\begin{bmatrix} e^{i\beta_2} - 1 & e^{i\alpha_2} - 1 & e^{i\gamma_2} - e^{i\gamma_1} \\ e^{i\beta_3} - 1 & e^{i\alpha_3} - 1 & e^{i\gamma_3} - e^{i\gamma_1} \\ e^{i\beta_4} - 1 & e^{i\alpha_4} - 1 & e^{i\gamma_4} - e^{i\gamma_1} \end{bmatrix} \begin{bmatrix} Z_9 \\ Z_{11} \\ Z_8 \end{bmatrix} = \begin{bmatrix} \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix}$$

Where γ is the angle of link Z_8 , β is the change in rotation of the link Z_9 and α is the change in rotation of link Z_{11} (free choices). Solving these equations, we get $Z_9 = 11.89$ cm and $Z_{11} = 26.11$ cm.

3.5 Summary

After multiple trial and errors and compiling all the results in a table, the link lengths are -

Link	Lengths (cm)
Z_1	15
Z_2	4.17
Z_3	20.33
Z_4	12
Z_5	20.33
Z_6	12
Z_7	11.14
Z_8	10
Z_9	11.89
Z_{10}	10
Z_{11}	26.11
Z_{12}	19

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4. ANALYSIS OF THE MECHANISM:

After deriving a solution in the previous section, it's essential to assess its validity and performance. Synthesis section already addressed this by applying constraints to the system to ensure its validity. However, additional analysis is needed to evaluate some constraints. Specifically, positional analysis is required to examine the motion of all links and the tracer path of the foot. Motion analysis is also necessary to estimate the acceptable range of the coupler angle (θ_c).

Previous research focused only on static forces, neglecting dynamic effects. To examine the dynamics, kinematic information for each link's direction, velocity, and acceleration over a cycle must be calculated first. Then, the system's dynamics can be determined using this information. However, deriving dynamic equations for each link presents a challenge due to the high number of unknowns, particularly regarding the reaction forces between the foot and the ground. The section will outline the kinematic equations to determine unknown angles, velocity, and acceleration for each link.

4.1 Kinematic Analysis

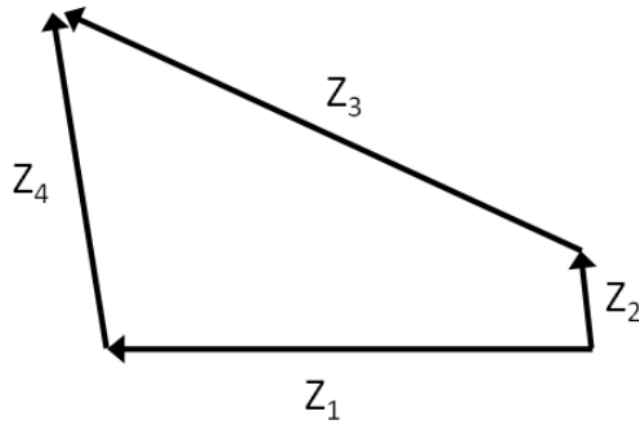
The positional analysis involves dividing the mechanism into three separate loops ($Z_1Z_2Z_3Z_4$, $Z_1Z_2Z_5Z_6$, and $Z_6Z_8Z_9Z_{10}$, as shown in Figure 3.1). By writing loop closure equations for each loop, the complete kinematic information for each link can be determined. However, to perform this analysis, the full kinematics of a single link must be known as it is a 1-DOF mechanism. In this case, the crank has a constant angular velocity, defining its kinematics. Therefore, kinematics of loop 1 ($Z_1Z_2Z_3Z_4$) & loop 2 ($Z_1Z_2Z_5Z_6$) must be solved first before addressing loop 3 ($Z_6Z_8Z_9Z_{10}$). The detailed derivations are provided in the subsequent sections.

4.1.1 Kinematic Loop 1 – $Z_1Z_2Z_3Z_4$

$$Z_1e^{i\theta_1} + Z_4e^{i\theta_4} = Z_2e^{i\theta_2} + Z_3e^{i\theta_3}$$

In this loop, we know the length of each link and we also have an angular position of link-1 (which is zero degree because fixed link) & link-2 (input). So, there are two unknowns in this loop, that are the angular position of link-3 & link-4 which we will find by splitting the above equation into imaginary and real components. Therefore, we get two equations and two variables to compute.

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Kinematic Loop-1

After splitting the vector loop equation in imaginary and real components, we get:

$$Z_1 \cos \theta_1 + Z_4 \cos \theta_4 = Z_2 \cos \theta_2 + Z_3 \cos \theta_3$$

$$Z_1 \sin \theta_1 + Z_4 \sin \theta_4 = Z_2 \sin \theta_2 + Z_3 \sin \theta_3$$

Here, we have two equations and two variables for θ_3 & θ_4 . Solving these equations, we get:

$$\theta_3 = \sin^{-1} \left[\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right]$$

where,

$$a = 4Z_3^2(Z_2^2 - 2Z_1Z_2 \cos(\theta_1 - \theta_2) + Z_1^2)$$

$$b = 4Z_3(Z_2 \sin \theta_2 - Z_1 \sin \theta_1)(Z_2^2 - 2Z_1Z_2 \cos(\theta_1 - \theta_2) + Z_1^2 + Z_3^2 - Z_4^2)$$

$$c = (Z_2^2 - 2Z_1Z_2 \cos(\theta_1 - \theta_2) + Z_1^2 + Z_3^2 - Z_4^2)^2 - 4Z_3^2(Z_2^2 \cos^2 \theta_2 - 2Z_1Z_2 \cos \theta_1 \cos \theta_2 + Z_1^2 \cos^2 \theta_1)$$

and,

$$\theta_4 = \cos^{-1} \left[\frac{Z_2 \cos \theta_2 - Z_1 \cos \theta_1 + Z_3 \cos \theta_3}{Z_4} \right]$$

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Now that we have all values of angular position and lengths of each link, we can derive the angular velocities and acceleration values for all the links by taking a derivative and double derivative of the kinematic loop respectively.

For angular velocity values,

$$\dot{\theta}_4 i Z_4 e^{i\theta_4} = \dot{\theta}_2 i Z_2 e^{i\theta_2} + \dot{\theta}_3 i Z_3 e^{i\theta_3}$$

Separating the above equation into its imaginary component and rearranging the equation,

$$\dot{\theta}_4 = \frac{\dot{\theta}_2 Z_2 \sin\theta_2 + \dot{\theta}_3 Z_3 \sin\theta_3}{Z_4 \sin\theta_4}$$

Now, separating the loop equation into real component, substituting the above equation and rearranging the equation, we get,

$$\dot{\theta}_3 = \frac{\dot{\theta}_2 Z_2 (\sin\theta_2 - \tan\theta_4 \cos\theta_2)}{Z_3 (\cos\theta_3 \tan\theta_4 - \sin\theta_3)}$$

Now we take the double derivative of the original kinematic loop equation to solve for the angular acceleration values of the links in the loop-1,

$$Z_4 (-\ddot{\theta}_4^2 + \ddot{\theta}_4 i) e^{i\theta_4} = Z_2 (-\ddot{\theta}_2^2 + \ddot{\theta}_2 i) e^{i\theta_2} + Z_3 (-\ddot{\theta}_3^2 + \ddot{\theta}_3 i) e^{i\theta_3}$$

Again after splitting the equation in imaginary and real components and solving the two equations, the final acceleration values we get are:

$$\ddot{\theta}_4 = \frac{A + Z_3 \ddot{\theta}_3 \sin\theta_3}{Z_4 \sin\theta_4}$$

where,

$$A = Z_2 (\dot{\theta}_2^2 \cos\theta_2 + \ddot{\theta}_2 \sin\theta_2) + Z_3 \dot{\theta}_3^2 \cos\theta_3 - Z_4 \dot{\theta}_4^2 \cos\theta_4$$

and,

$$\ddot{\theta}_4 = \frac{B + Z_3 \ddot{\theta}_3 \cos\theta_3}{Z_4 \cos\theta_4}$$

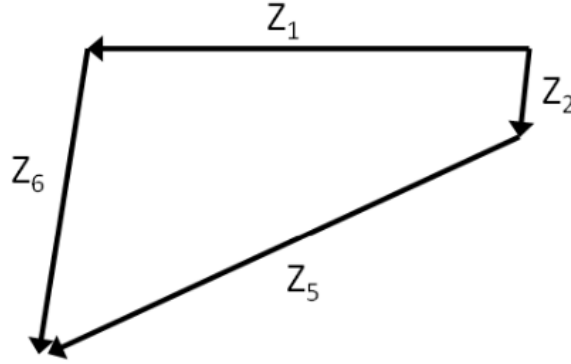
$$B = Z_2 (-\dot{\theta}_2^2 \sin\theta_2 + \ddot{\theta}_2 \cos\theta_2) - Z_3 \dot{\theta}_3^2 \sin\theta_3 + Z_4 \dot{\theta}_4^2 \sin\theta_4$$

$$\ddot{\theta}_3 = \frac{B \tan\theta_4 - A}{Z_3 (\sin\theta_3 - \cos\theta_3 \tan\theta_4)}$$

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Since we got the angular acceleration for link-3, we can substitute this value in the equation for link-4 to find the angular acceleration of link-4.

4.1.2 Kinematic Loop 2 – $Z_1 Z_2 Z_5 Z_6$



This loop is exactly the same as the previous loop in terms of length, but is in different orientation. So, the equations for all the angular values for link-5 & link-6 will not be derived again.

Angular position equations:

$$a = 4Z_5^2(Z_2^2 - 2Z_1Z_2 \cos(\theta_1 - \theta_2) + Z_1^2)$$

$$b = 4Z_5(Z_2 \sin \theta_2 - Z_1 \sin \theta_1)(Z_2^2 - 2Z_1Z_2 \cos(\theta_1 - \theta_2) + Z_1^2 + Z_5^2 - Z_6^2)$$

$$c = (Z_2^2 - 2Z_1Z_2 \cos(\theta_1 - \theta_2) + Z_1^2 + Z_5^2 - Z_6^2)^2 - 4Z_5^2(Z_2^2 \cos^2 \theta_2 - 2Z_1Z_2 \cos \theta_1 \cos \theta_2 + Z_1^2 \cos^2 \theta_1)$$

$$\theta_5 = \sin^{-1} \left[\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right]$$

$$\theta_6 = \cos^{-1} \left[\frac{Z_2 \cos \theta_2 - Z_1 \cos \theta_1 + Z_5 \cos \theta_5}{Z_6} \right]$$

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Angular velocity equations:

$$\dot{\theta}_6 = \frac{\dot{\theta}_2 Z_2 \sin \theta_2 + \dot{\theta}_5 Z_5 \sin \theta_5}{Z_6 \sin \theta_6}$$

$$\dot{\theta}_5 = \frac{\dot{\theta}_2 Z_2 (\sin \theta_2 - \tan \theta_6 \cos \theta_2)}{Z_5 (\cos \theta_5 \tan \theta_6 - \sin \theta_5)}$$

Angular acceleration equations:

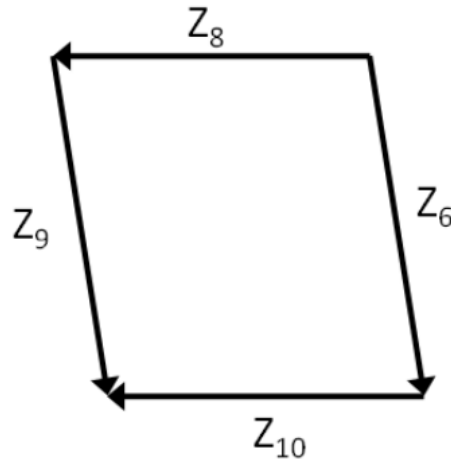
$$A = Z_2 (\dot{\theta}_2^2 \cos \theta_2 + \ddot{\theta}_2 \sin \theta_2) + Z_5 \dot{\theta}_5^2 \cos \theta_5 - Z_6 \dot{\theta}_6^2 \cos \theta_6$$

$$B = Z_2 (-\dot{\theta}_2^2 \sin \theta_2 + \ddot{\theta}_2 \cos \theta_2) - Z_5 \dot{\theta}_5^2 \sin \theta_5 + Z_6 \dot{\theta}_6^2 \sin \theta_6$$

$$\ddot{\theta}_5 = \frac{B \tan \theta_6 - A}{Z_5 (\sin \theta_5 - \cos \theta_5 \tan \theta_6)}$$

$$\ddot{\theta}_6 = \frac{B + Z_5 \ddot{\theta}_5 \cos \theta_5}{Z_6 \cos \theta_6}$$

4.1.3 Kinematic Loop 3 – $Z_6 Z_8 Z_9 Z_{10}$



In this kinematic loop, we are using the same method to solve all the unknown variables as we used for the previous loops. For that, we know the lengths of all the links, we found the angular position, velocity & acceleration for link-6 in the previous loops. And, Since, link-8, 4 & 7

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together make a rigid link, we can get the relations for the same between link-8 & link-4 whose values we already have.

After, deriving all the equations for angular parameters for link-9 & link-10 using the same method we used for previous loops, we get:

Angular position equations:

$$Z_8 e^{i\theta_8} + Z_9 e^{i\theta_9} = Z_6 e^{i\theta_6} + Z_{10} e^{i\theta_{10}}$$

$$a = 4Z_{10}^2(Z_6^2 - 2Z_8Z_6 \cos(\theta_8 - \theta_6) + Z_8^2)$$

$$b = 4Z_{10}(Z_6 \sin \theta_6 - Z_8 \sin \theta_8)(Z_6^2 - 2Z_8Z_6 \cos(\theta_8 - \theta_6) + Z_8^2 + Z_{10}^2 - Z_9^2)$$

$$c = (Z_6^2 - 2Z_8Z_6 \cos(\theta_8 - \theta_6) + Z_8^2 + Z_{10}^2 - Z_9^2)^2 - 4Z_{10}^2(Z_6^2 \cos^2 \theta_6 - 2Z_8Z_6 \cos \theta_8 \cos \theta_6 + Z_8^2 \cos^2 \theta_8)$$

$$\theta_{10} = \sin^{-1} \left[\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right]$$

$$\theta_9 = \cos^{-1} \left[\frac{Z_6 \cos \theta_6 - Z_8 \cos \theta_8 + Z_{10} \cos \theta_{10}}{Z_9} \right]$$

Angular velocity equations:

$$\dot{\theta}_9 = \frac{\dot{\theta}_6 Z_6 \sin \theta_6 + \tan \theta_{10} (\dot{\theta}_8 Z_8 \cos \theta_8 - \dot{\theta}_6 Z_6 \cos \theta_6) - \dot{\theta}_8 Z_8 \sin \theta_8}{Z_9 (\sin \theta_{10} - \cos \theta_9 \tan \theta_{10})}$$

$$\dot{\theta}_{10} = \frac{\dot{\theta}_8 Z_8 \cos \theta_8 - \dot{\theta}_6 Z_6 \cos \theta_6 + \dot{\theta}_9 Z_9 \cos \theta_9}{Z_{10} \cos \theta_{10}}$$

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Angular acceleration equations:

$$\begin{aligned} Z_8(-\dot{\theta}_8^2 + \ddot{\theta}_8 i)e^{i\theta_8} + Z_9(-\dot{\theta}_9^2 + \ddot{\theta}_9 i)e^{i\theta_9} \\ = Z_6(-\dot{\theta}_6^2 + \ddot{\theta}_6 i)e^{i\theta_6} + Z_{10}(-\dot{\theta}_{10}^2 + \ddot{\theta}_{10} i)e^{i\theta_{10}} \end{aligned}$$

$$\ddot{\theta}_{10} = \frac{B \tan \theta_9 - A}{Z_{10}(\sin \theta_{10} - \cos \theta_{10} \tan \theta_9)}$$

$$\ddot{\theta}_9 = \frac{B + Z_{10} \ddot{\theta}_{10} \cos \theta_{10}}{Z_9 \cos \theta_9}$$

$$\begin{aligned} A = Z_6(\dot{\theta}_6^2 \cos \theta_6 + \ddot{\theta}_6 \sin \theta_6) + Z_{10} \dot{\theta}_{10}^2 \cos \theta_{10} - Z_8(\dot{\theta}_8^2 \cos \theta_8 + \\ \ddot{\theta}_8 \sin \theta_8) - Z_9 \dot{\theta}_9^2 \cos \theta_9 \end{aligned}$$

$$\begin{aligned} B = Z_6(-\dot{\theta}_6^2 \sin \theta_6 + \ddot{\theta}_6 \cos \theta_6) - Z_{10} \dot{\theta}_{10}^2 \sin \theta_{10} - Z_8(-\dot{\theta}_8^2 \sin \theta_8 + \\ \ddot{\theta}_8 \cos \theta_8) + Z_9 \dot{\theta}_9^2 \sin \theta_9 \end{aligned}$$

5. DESIGN ITERATIONS USING KINEMATIC ANALYSIS & COUPLER CURVE SIMULATION

From the derived equations for analysis the link lengths are checked for the following constraints.

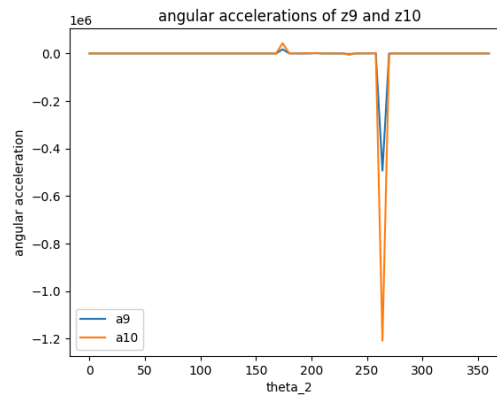
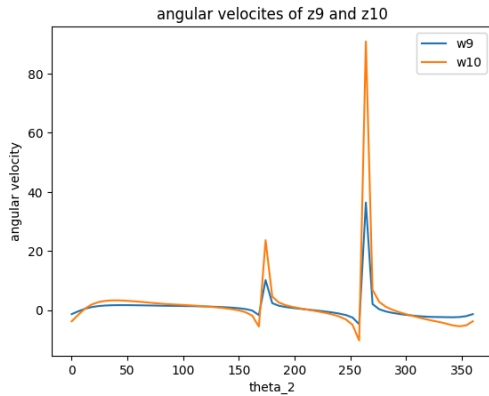
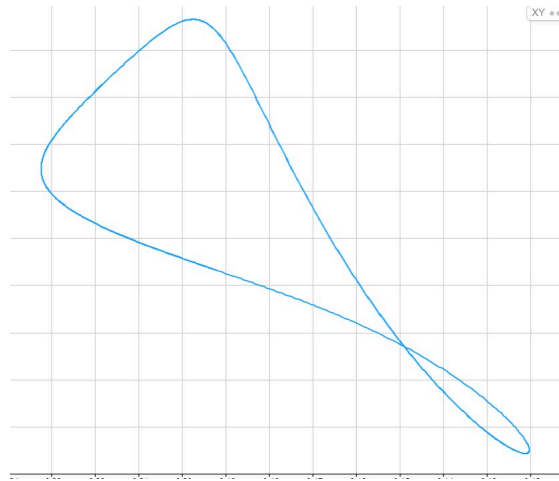
- The ankle of this mechanism should follow a walking motion coupler curve with a flat stride touching the ground.
- Grashoff criterion should be satisfied for the appropriate four bars present in this mechanism (kinematic loop-1 & loop-2).
- Smoothen out the kinematic properties of all the links to reduce sudden jerks in the mechanism.

From the literature survey we conducted, appropriate free choices were made. According to the consequent analysis, and trial and error, the free choices were updated. Three of these iterations are as mentioned below.

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5.1 Iteration-I

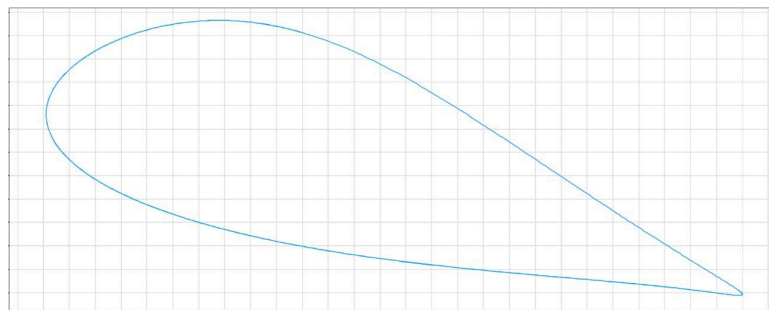
Link	Lengths (cm)
Z_1	15
Z_2	3.94
Z_3	23.279
Z_4	14.102
Z_5	23.279
Z_6	12.378
Z_7	12.376
Z_8	5.047
Z_9	12.387
Z_{10}	5.047
Z_{11}	10.669
Z_{12}	10.669



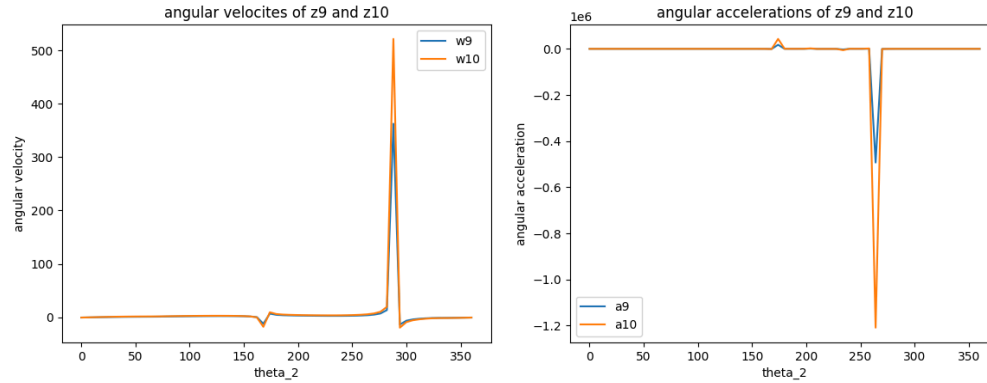
It doesn't satisfy the first constraint of stride. Also, the jerk values are extremely high. Thus, another iteration is required.

5.2 Iteration-II

Link	Lengths (cm)
Z_1	15
Z_2	2.71
Z_3	17.86
Z_4	6.99
Z_5	17.86
Z_6	10.079
Z_7	6.99
Z_8	7.0089
Z_9	10.0789
Z_{10}	7.008
Z_{11}	11.561
Z_{12}	11.561



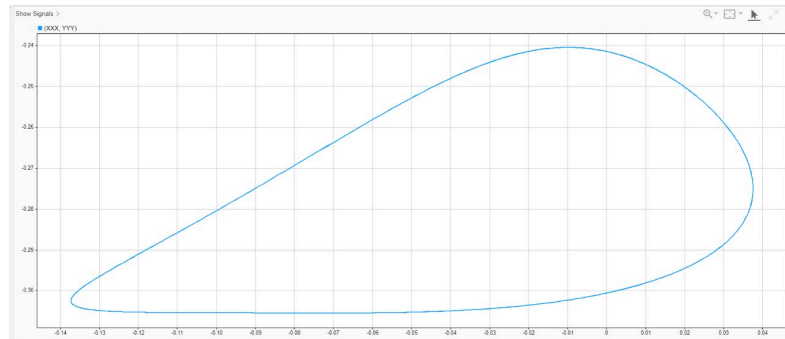
SYNTHESIS & ANALYSIS OF THEO JANSEN MECHANISM



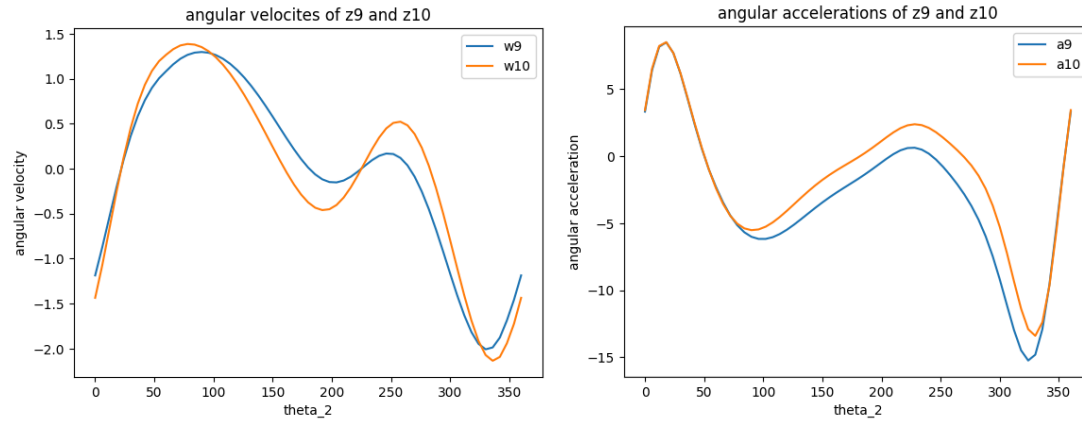
After trial and error, we arrived at new link lengths values whose coupler curve satisfied the first constraint. But, it was not horizontal. As the jerks still have high values, another iteration is required.

5.3 Iteration-III

Link	Lengths (cm)
Z_1	15
Z_2	4.17
Z_3	20.33
Z_4	12
Z_5	20.33
Z_6	12
Z_7	11.14
Z_8	10
Z_9	11.89
Z_{10}	10
Z_{11}	26.11
Z_{12}	19



SYNTHESIS & ANALYSIS OF THEO JANSEN MECHANISM



The mechanism satisfies all the conditions.

Grashof criterion for the kinematic loop-1 & loop-2:

$$s + l < p + q$$

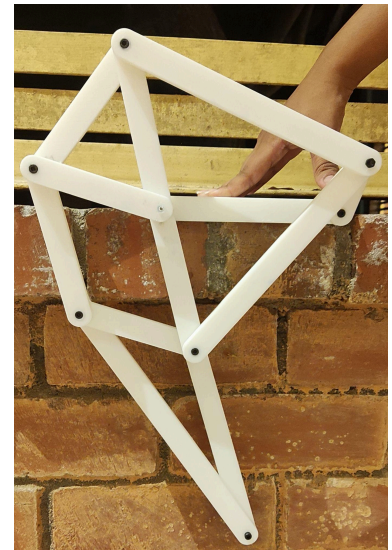
Where, for loop-1 $\Rightarrow s = 4.17$; $l = 20.33$; $p = 15$ & $q = 12$

And for loop-2 $\Rightarrow s = 4.17$; $l = 20.33$; $p = 15$ & $q = 12$

The angular velocities and accelerations vary smoothly with minimum jerks.

6. DESIGN & FABRICATION:

A CAD model designed using the Fusion 360 software was used to 3D print the mechanism using PLA plastic in the Central Workshop. Nuts and Bolts were used as revolute joints in the mechanism.



7. REFERENCES:

Suraj Jaiswal "Simulation of a Mechatronic Machine" at LUT University, Lappeenranta, Finland-[Url](#)

Kinematic simulation, by mechverse -[link to video](#)

Mathwork documentation on multibody simscape

SYNTHESIS & ANALYSIS OF THEO JANSEN MECHANISM

“Design and Optimization of a one degree of freedom eight bar mechanism for walking machine”, author Daniel Giesbrecht

OpenSCAD User Manual/Example/Strandbeest

“Design and Linkage Analysis of Theo Jansen Mechanism” by Yash Punde, Yugandhar Dhande, Amit Chopde, Sumedh Bagade

“Study of Theo Jansen Walking Machine” Author Shubham R. Kurkure, Vaibhav R. Patil, Kalpesh P. Khute, Akshay Patil, Lalit Zalte, Prof. Mayur P. Thakur